data-analysis-worksheet

Data Analysis Challenge, Worksheet

Exercise 3: AI-Powered Data Analysis (1:45 PM - 2:30 PM)

Your dataset: `retailflow-customer-service-data.csv` **What it contains:** 260 customer service tickets from RetailFlow over 6 months (July–December 2024) **Columns:** ticket_id, date, day_of_week, month, category, channel, priority, agent, response_time_hours, resolution_time_hours, satisfaction_score, resolved

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How This Exercise Works

You'll use the **AI-Assisted Analysis Workflow** (Prepare → Upload → Question → Verify → Refine → Present) to answer three business questions. Work individually or in pairs.

Phase	Time	What You Do
Upload & orient	5 min	Get the data into your AI tool, ask it to describe what it sees
Questions 1 & 2	15 min	Work through the first two questions using progressive prompting
Verification Challenge	10 min	Test three "insight claims" against the actual data, trust nothing on fluency alone
Question 3 & your brief	10 min	Form recommendations, then have AI format your one-page brief
Debrief	10 min	Compare findings and claim verdicts with other participants

(Work through the steps in order, the Challenge sits between Question 2 and Question 3 on purpose: audit what you have before you build recommendations on top of it. The debrief may run a few minutes into afternoon tea.)

Step 1: Prepare (Before You Prompt)

Before opening your AI tool, think about what you're looking for.

The situation: RetailFlow's customer service manager, Karen, has asked you to review 6 months of ticket data and give her three actionable insights. She doesn't want raw numbers, she wants to know what's going wrong and what to do about it.

Your questions (read all three before starting):

1. What are the top 3 drivers of customer dissatisfaction?
2. Are there patterns in when response times blow out?
3. What would you recommend to the customer service manager?

Step 2: Upload & Orient

Upload the CSV to your AI tool (ChatGPT, Claude, or Gemini). **Start a fresh chat for this**: long conversations drift toward your view, and for this exercise you want an independent read on the data.

Orientation prompt (use this to start):

I'm uploading a CSV file of customer service ticket data from RetailFlow, a retail company. It contains 260 tickets over 6 months (July–December 2024). Please describe the dataset: how many rows, what columns exist, the range of dates, and a brief summary of what you see.

What the AI told you about the dataset:

Does this match what you expected? (This is your first verification step.)

☐ Yes, looks right

☐ Something seems off, I need to clarify

Step 3: Question, Business Question 1

What are the top 3 drivers of customer dissatisfaction?

Start broad, then go deeper.

Prompt 1 (broad):

What is the average satisfaction score overall? Now break it down by category. Which categories have the lowest satisfaction?

Findings:

Prompt 2 (deeper):

For the lowest-satisfaction categories, what else do you notice? Look at response times, resolution rates, and channels for those categories compared to the higher-satisfaction ones.

Findings:

Prompt 3 (causal):

Is there a relationship between response time and satisfaction score across all tickets? What happens to satisfaction when response time exceeds 8 hours? What about 12 hours?

Findings:

Your answer to Question 1, Top 3 drivers of dissatisfaction:

#	Driver	Evidence
1		
2		
3		

Step 4: Question, Business Question 2

Are there patterns in when response times blow out?

Prompt 1 (by day):

What is the average response time by day of the week? Which days are worst?

Findings:

Prompt 2 (by month):

What is the average response time by month? Is there a seasonal pattern?

Findings:

Prompt 3 (by channel):

What is the average response time by channel? Which channels are fastest and slowest?

Findings:

Prompt 4 (combining factors):

What happens when the worst day, worst month, and slowest channel overlap? How bad does it get?

Findings:

Your answer to Question 2, When do response times blow out?

Pattern	Detail	How Bad?
Day of week		
Month/season		
Channel		
Worst combination		

Step 5: Verify

Before moving to recommendations, check the AI's work. Fluency is not evidence, a fluent, confident, professional-sounding finding and a correct one look identical until someone checks. The AI cannot reliably run that check on itself. You, with the actual data, can.

Pick 2-3 specific claims the AI made and verify them:

AI Claimed	I Checked	Correct?
		Yes / No / Close enough
		Yes / No / Close enough
		Yes / No / Close enough

Verification methods: - Sort/filter the CSV in a spreadsheet and spot-check a number - Ask the AI to "show me the raw data for [specific claim]" - Use common sense, does this finding make sense given what you know about customer service?

Did the AI get anything wrong?

Step 5.5: The Verification Challenge

Three "insights" about the RetailFlow data are circulating. Each one sounds plausible and professional. **At least one is wrong, and at least one is stated wrong.** Your job: verify each claim against the actual CSV and classify it.

This is the core skill of the whole day. The claims below stand in for every AI output you'll ever be tempted to forward: the danger isn't obvious nonsense, it's the convincing average, output that carries all the form of good analysis without the substance. The only defence is a check from outside.

How to verify: use your AI tool, a spreadsheet filter, or both, but the ruling comes from the data, not from how confident the claim sounds. *(Asking the AI "is this right?" doesn't count, the model will happily reaffirm its own claim. Check against the data itself.)*

The three claims

Claim A: > "Email is roughly three times slower than Live Chat."

Claim B: > "Account Issues has the lowest satisfaction of any category, which makes sense, because account problems are the most complex to resolve."

Claim C: > "Sunday response times are more than double the overall average."

Your verdicts

Claim	My verdict (circle one)	Evidence from the data
A	Verified / Refuted / Overstated	
B	Verified / Refuted / Overstated	
C	Verified / Refuted / Overstated	

Verdict key: Verified = accurate as stated · **Refuted** = the data contradicts it · **Overstated** = directionally right, but the specific number is wrong

Which claim would have been most damaging to forward to Karen unverified? Why?

Trust tool check: Claim C turns on a precise number that would drive a staffing decision. That's the bottom half of the trust grid, precision required. "About right" is the same as wrong there, no matter how good the sentence sounds.

Step 6: Question, Business Question 3

What would you recommend to the customer service manager?

Now that you have the data, ask the AI to help form recommendations.

Prompt:

Based on everything we've found, what are your top 3 recommendations for Karen, the customer service manager at RetailFlow? Each recommendation should: (1) state the problem it addresses, (2) describe the specific action, and (3) explain the expected impact. Prioritise by potential impact on customer satisfaction.

AI's recommendations:

Your refined recommendations (edit the AI's suggestions with your own judgment):

#	Recommendation	Why	Expected Impact
1			
2			
3			

Step 7: Present

Have AI format your findings as a one-page brief.

Prompt:

Create a one-page insight brief for Karen, the RetailFlow customer service manager. Include: - A 2-sentence executive summary - Top 3 drivers of customer dissatisfaction (with data) - Key response time patterns (with data) - 3 prioritised recommendations

Format with clear headers and bullet points. Keep it under 400 words. Use a professional but direct tone, Karen is busy and wants actionable insights, not a data dump.

Paste or save your final brief here (or print it):

Debrief Questions

Discuss at your table:

1. Did everyone's AI find the same patterns? - [ ] Yes, very similar findings - [ ] Mostly similar, some differences - [ ] Significant differences

2. Where did the AI analysis surprise you?

3. In the Verification Challenge, which claims did you classify correctly, and which one fooled you? What made it convincing?

4. How long would this analysis have taken without AI?

5. Would you trust this analysis enough to present it to a real manager? - [ ] Yes, as-is - [ ] Yes, with the verification I did - [ ] Only as a starting point, I'd need to dig deeper

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Reflection

What did you learn about using AI for data analysis?

What would you do differently next time?